Databuses: MIL-STD-1553

CEC 300: Computing in Aviation and Aerospace

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Outline

- Background
- Physical Layer
  - Hardware
  - Signals
- Components
  - Bus Controller
  - Bus Monitor
  - Remote Terminal
- Redundancy
- Message Types and Formats

Background

- First adopted by the US Air Force in 1973
  - Version B adopted 1978
  - Version C adopted 2018

- Applications in military aircraft, civilian aircraft, and spacecraft
  - Spacecraft include the James Webb Telescope and the International Space Station

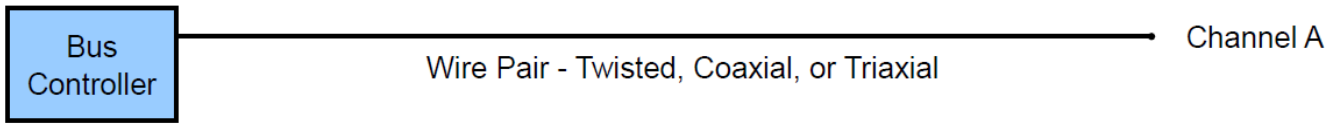
- Topology: Bus

- Key Components:
  - Transmission line(s)
  - Bus Controller
  - Remote Terminals (up to $2^5 - 2$ or 30 remote terminals or monitors)
  - Bus Monitor

Link to Standard Version B: nepp.nasa.gov/docuploads/43745C0A-323E-4346-A434F4342178CD0E/MIL-STD-1553.pdf

Physical Layer

MIL-STD-1553B Avionic Data Bus



Physical Layer Characteristics

(Signals are measured differentially, line to line)

Output	**Transformer Coupled**	**Direct Coupled**
Bit Rate	1 mb/s	1 mb/s
Output Level, (volts)	18.0 min, 27.0 max	6.0 min, 9.0 max
Rise time, fall time	100 ns, min, 300 ns, max	100 ns, min, 300 ns, max
Distortion	900 mv, peak	300 mv, peak
DC offset	250 mv, max	90 mv, max
Input		
Input Level, (volts)	0.86 min, 14.0 max	1.2 min, 20 max
Noise Immunity	0.2 volts	0.28 volts
Com. Mode Rejection	10 volts peak	10 volts peak
Input Impedance	1000 ohms	2000 ohms

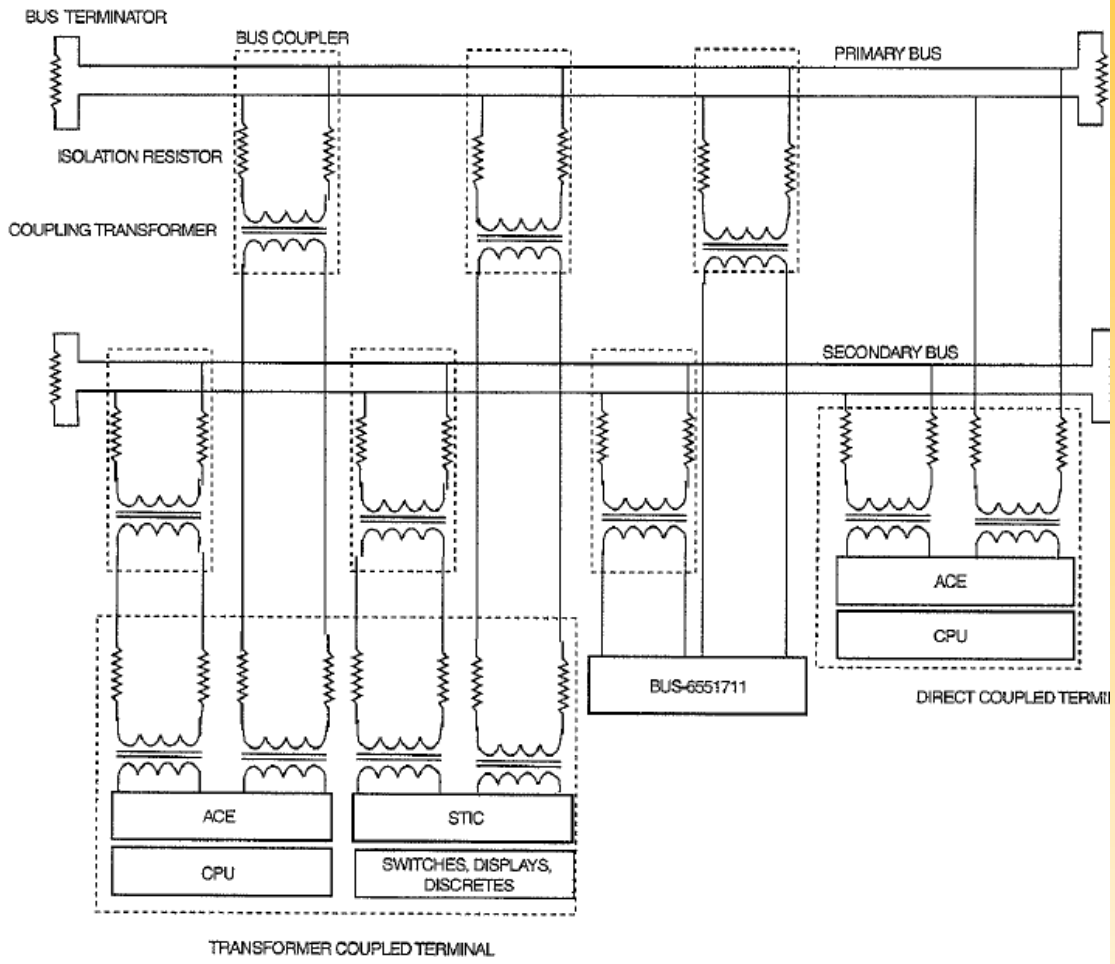
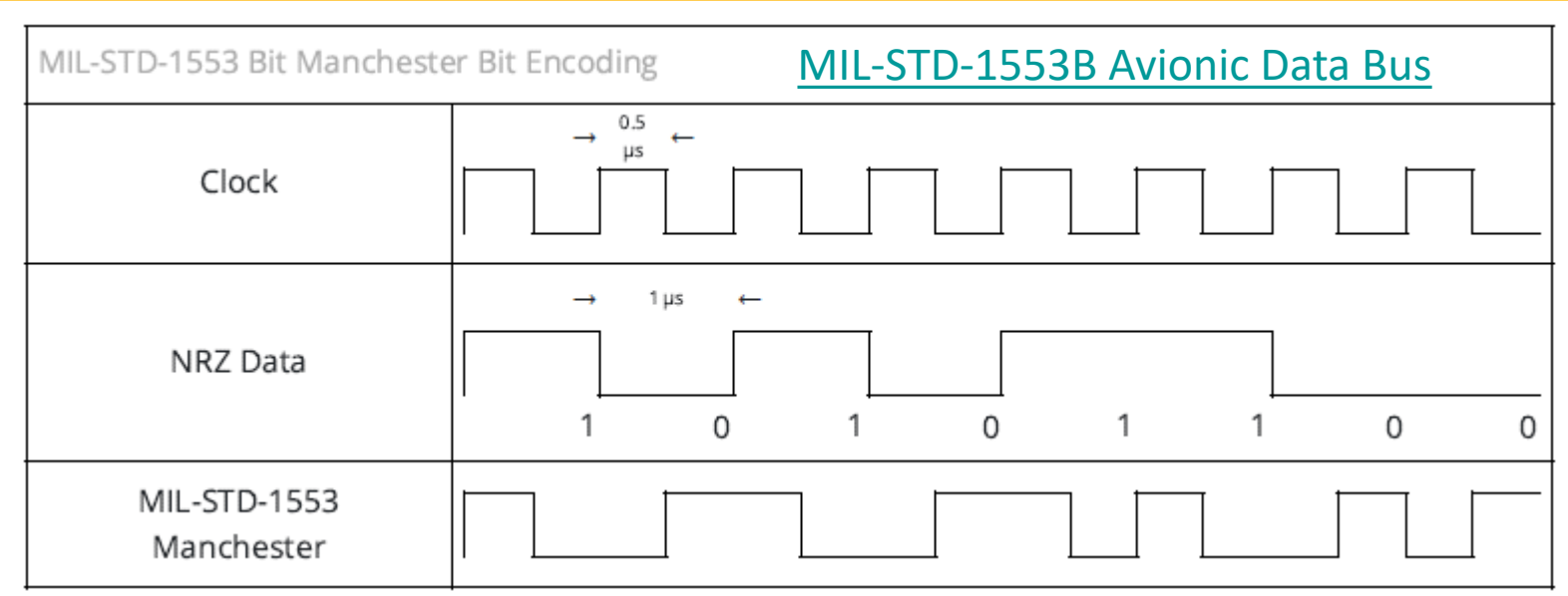
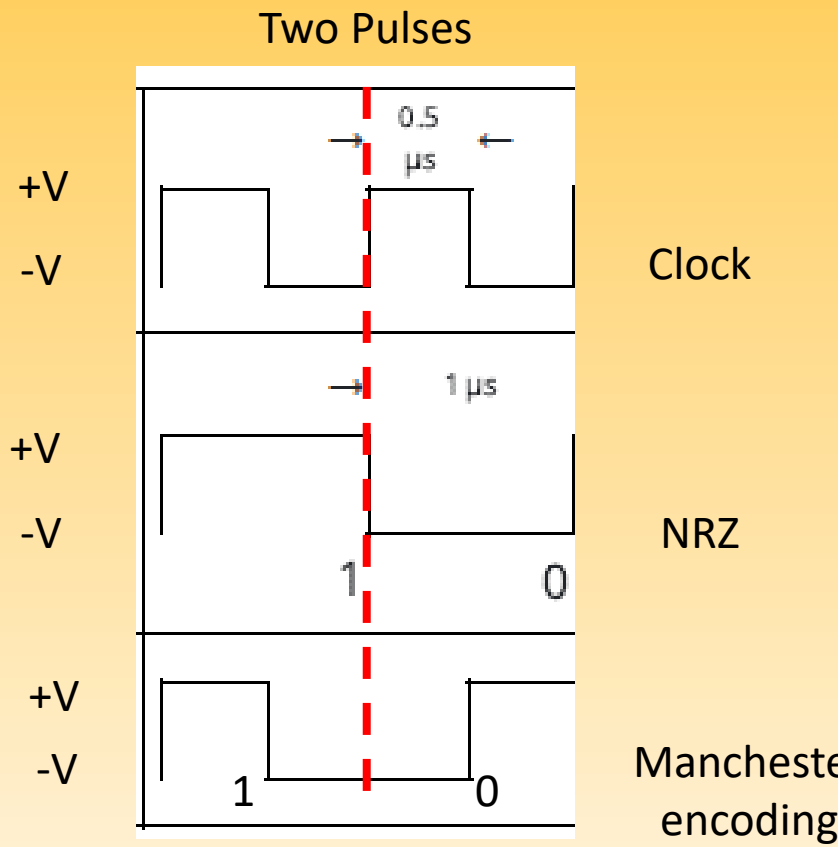


Figure 7.10 MIL STD 1553 communications system

(Helfrick, 2007)

Physical Layer and Manchester Encoding





- Bus cable / transmission line – coaxial or twisted-pair
- Transformer couplers for (1) connection of remote terminal to stub and (2) cable extension.
- Manchester encoding
  - Codes signal such that regardless of the number of 1s or 0s, the average voltage will remain at zero.
  - Easy to discern logic value

Components

Mapping of Function to Components

Table I-4.1 Bus Elements Capabilities

Capability	Bus controller	Minimum standalone RT (dual bus)	Standalone RT (dual bus)	Standalone RT (multiple bus)	Minimum embedded terminal (dual bus)	Embedded terminal (dual bus)	Embedded terminal (multiple bus)	Intelligent device—RT or embedded (dual bus)	Intelligent device—RT or embedded (multiple bus)
1. Information transfer formats									
a. Controller to remote terminal	X	X	X	X	X	X	X	X	X
b. Remote terminal to controller	X	X	X	X	X	X	X	X	X
c. Remote terminal to remote terminal	X	X	X	X	X	X	X	X	X
d. Mode command without data word	X		X	X		X	X	X	X
e. Mode command with data word (transmit)	X		X	X		X	X	X	X
f. Mode command with data word (receive)	X			X			X	X	X
2. Broadcast information transfer formats									
a. Controller to RT(s) transfer	X					X	X	X	X
b. RT to RT(s) transfer	X					X	X	X	X
c. Mode command without data word	X		X	X		X	X	X	X
d. Mode command with data word	X		X	X			X	X	X

Capability	Bus controller	Minimum standalone RT (dual bus)	Standalone RT (dual bus)	Standalone RT (multiple bus)	Minimum embedded terminal (dual bus)	Embedded terminal (dual bus)	Embedded terminal (multiple bus)	Intelligent device—RT or embedded (dual bus)	Intelligent device—RT or embedded (multiple bus)
3. Mode codes									
a. Dynamic bus control	X								
b. Synchronize	X		X	X					
c. Transmit status word	X	X	X	X	X	X	X	X	X
d. Initiate self-test	X		X	X					
e. Transmitter shutdown	X		X			X		X	
f. Override transmitter shutdown	X		X			X		X	
g. Inhibit terminal flag bit	X		X	X				X	X
h. Override inhibit terminal flag bit	X		X	X				X	X
i. Reset remote terminal	X		X	X		X	X		X
j. Transmit vector word	X		X	X			X	X	X
k. Synchronize	X							X	X
l. Transmit last command	X		X	X		X	X	X	X
m. Transmit bit word	X		X	X			X		X
n. Selected transmitter shutdown	X			X					X
o. Override selected transmitter shutdown	X			X					X
4. Status bit field									
a. Message error		X	X	X	X	X	X	X	X
b. Instrumentation (set to zero)									
c. Service Request			X	X			X	X	X
d. Broadcast received command			X	X	X	X	X	X	X
e. Busy			X	X		X	X	X	X
f. Subsystem flag						X	X		
g. Dynamic bus control acceptance								X	X
h. Terminal flag			X	X				X	X

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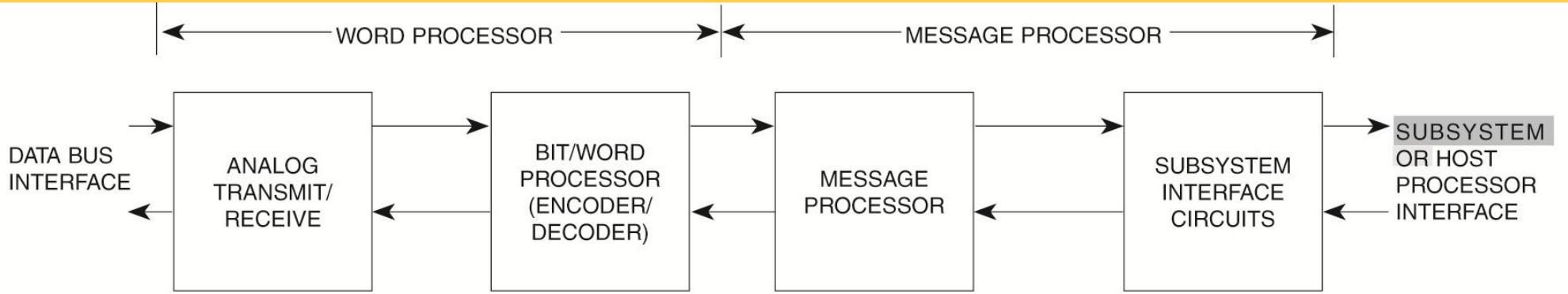
Component: Bus Controller / Bus Monitor



- Receive
  - Signal limiting
  - Filter
  - Threshold detection
- Transmit
  - Driver
  - Transmitter control
  - Timer
- Common
  - Bus coupling

- Receive
  - Receiver
  - Sync detection
  - Data detection
  - Manchester error detection
  - Parity check
  - Bits/word count
- Transmit
  - Transmit control
  - Sync/data encode
  - Parity generation
- Clock generation

- Receive
  - Command word decode
  - Status word decode
  - Address recognition
  - Mode execution
  - Word count recognition
  - Message error detection
- Transmit
  - Word count (BC)
  - Status register
- Built-in test
- Subsystem interface control
  - Data address
  - Control registers
- Memory buffer

- Channel selection
- Data sampling
- Conversion
- Subsystem timing
- Buffer registers
- Calibration
- Self-test
- DMA to host memory
- Interrupt lines
- Control lines

BC = Bus controller

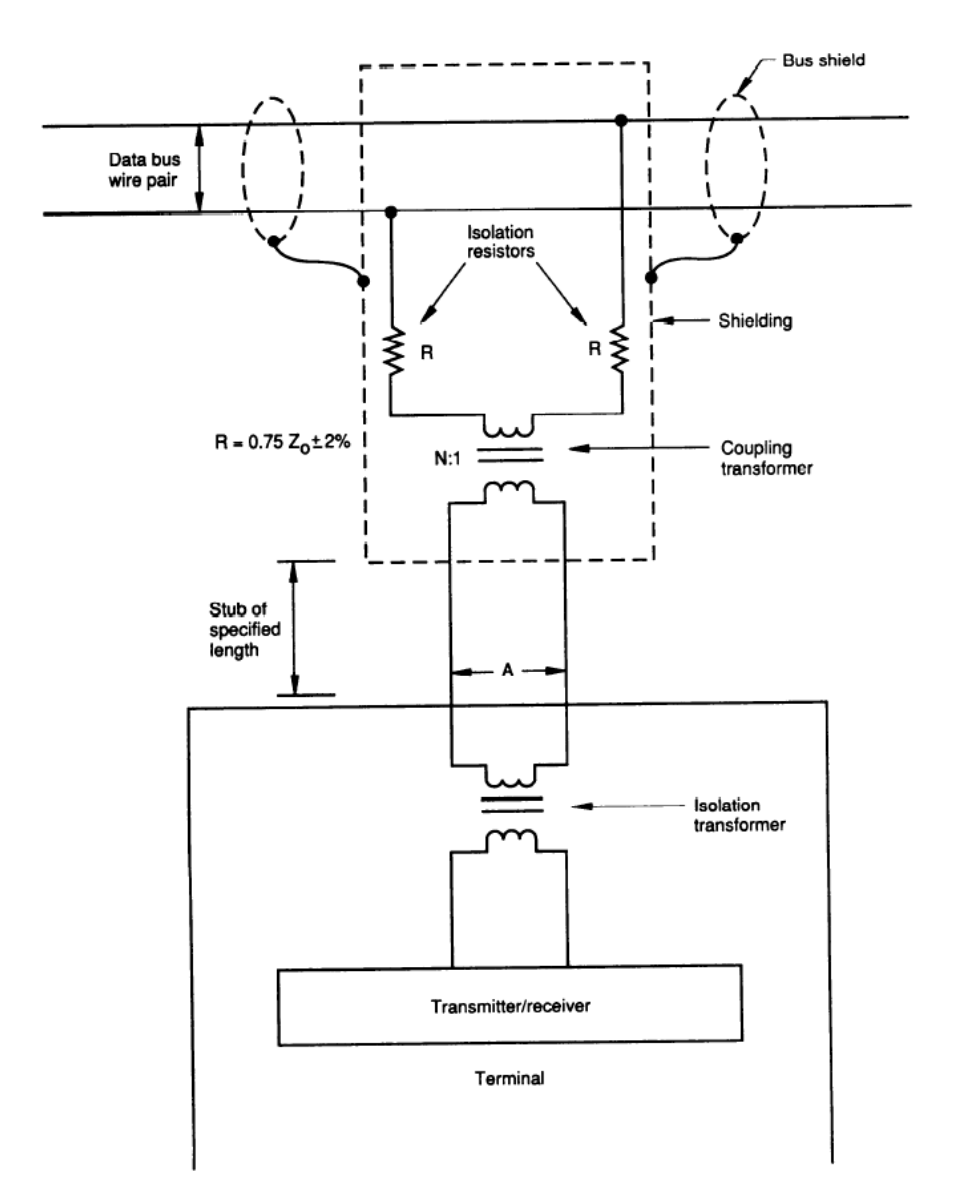
Not used in bus monitor

4 - Hardware - MIL-STD-1553

Component: Remote Terminals

- Remote Terminals
  - Terminals that are not a bus controller or monitor

- Coupling to bus:
  - Transformer Coupling
    - Long-Stub > 1ft
    - Isolation of main bus from faults on stub
    - Reduce signal distortion by increasing stub's effective impedance
  - Direct Coupling
    - Short-Stub < 1ft



MIL-HDBK-1553A
Transformer Coupling

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Scheduler

- Discussed in later video.

- Integrated into the bus controller
  - Enables deterministic, time-division multiplexing
  - Pre-determined priority schema
  - Frame and sub-frame time scheduling

- Major frames can contain multiple minor frames
  - Supports both periodic and aperiodic messages

- Scheduling mechanism is not prescribed. Up to the developer.

Redundancy

- To address the risk of failure, the network can use redundancy to mitigate the risk of system failure.
- The figure shows the addition of a backup or fallback bus controller and a fallback channel.
- Terminals and monitors are connected to both the primary and secondary bus (channels).
- If the primary bus or its controller fails, the system can identify the fault and switch to the fallback controller and bus.

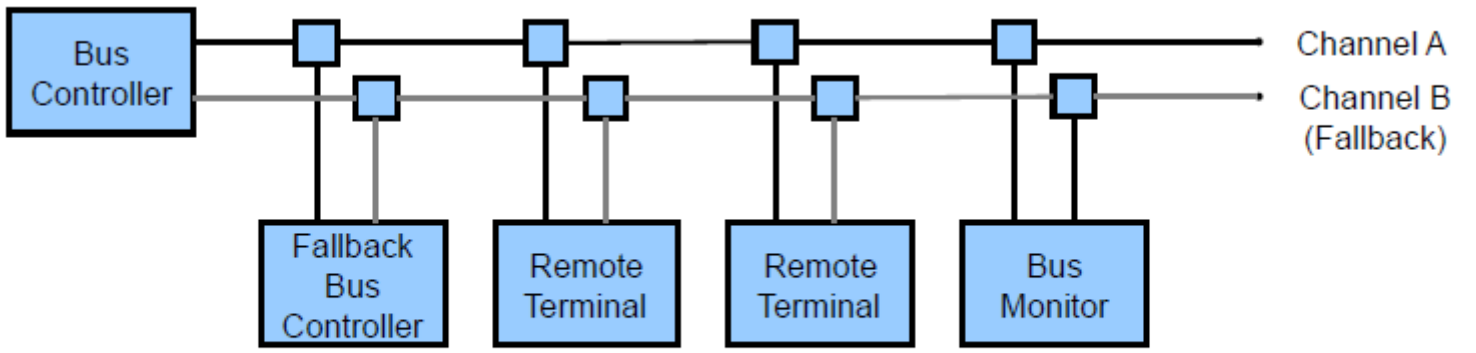

Fig. 4 - MIL-STD-1553 Bus Topology Part 4

MIL-STD-1553B Avionic Data Bus

Also, while not redundancy, it should be noted a systems using MIL-STD-1553 can utilize one or multiple buses.

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Message Formats

Information Transfer Messages

Bus Controller to Remote Terminal Transfers								
RX Command	Data Word	Data Word	•••••	Data Word	• •	Status Word	#	Next Comm-and

Mode Command with Data Word (TX)					
Mode Comm-and	• •	Status Word	Data Word	#	Next Comm-and

Remote Terminal to Bus Controller Transfers								
TX Command	• •	Status Word	Data Word	Data Word	•••••	Data Word	#	Next Comm-and

Mode Command without Data Word				
Mode Comm-and	• •	Status Word	#	Next Comm-and

Remote Terminal to Remote Terminal Transfers										
RX Command	TX Command	• •	Status Word	Data Word	•••••	Data Word	• •	Status Word	#	Next Comm-and

Mode Command with Data Word (RX)					
Mode Comm-and	Data Word	• •	Status Word	#	Next Comm-and

Broadcast Messages

Bus Controller to Remote Terminal Broadcast						
RX Command	Data Word	Data Word	• •	Data Word	#	Next Comm-and

Remote Terminal to Remote Terminal Broadcast								
RX Command	TX Command	• •	Status Word	Data Word	•••••	Data Word	#	Next Comm-and

Mode Command with Data Word (RX)			
Mode Comm-and	Data Word	#	Next Comm-and

Mode Command without Data Word		
Mode Comm-and	#	Next Comm-and

MIL-STD-1553B Avionic Data Bus

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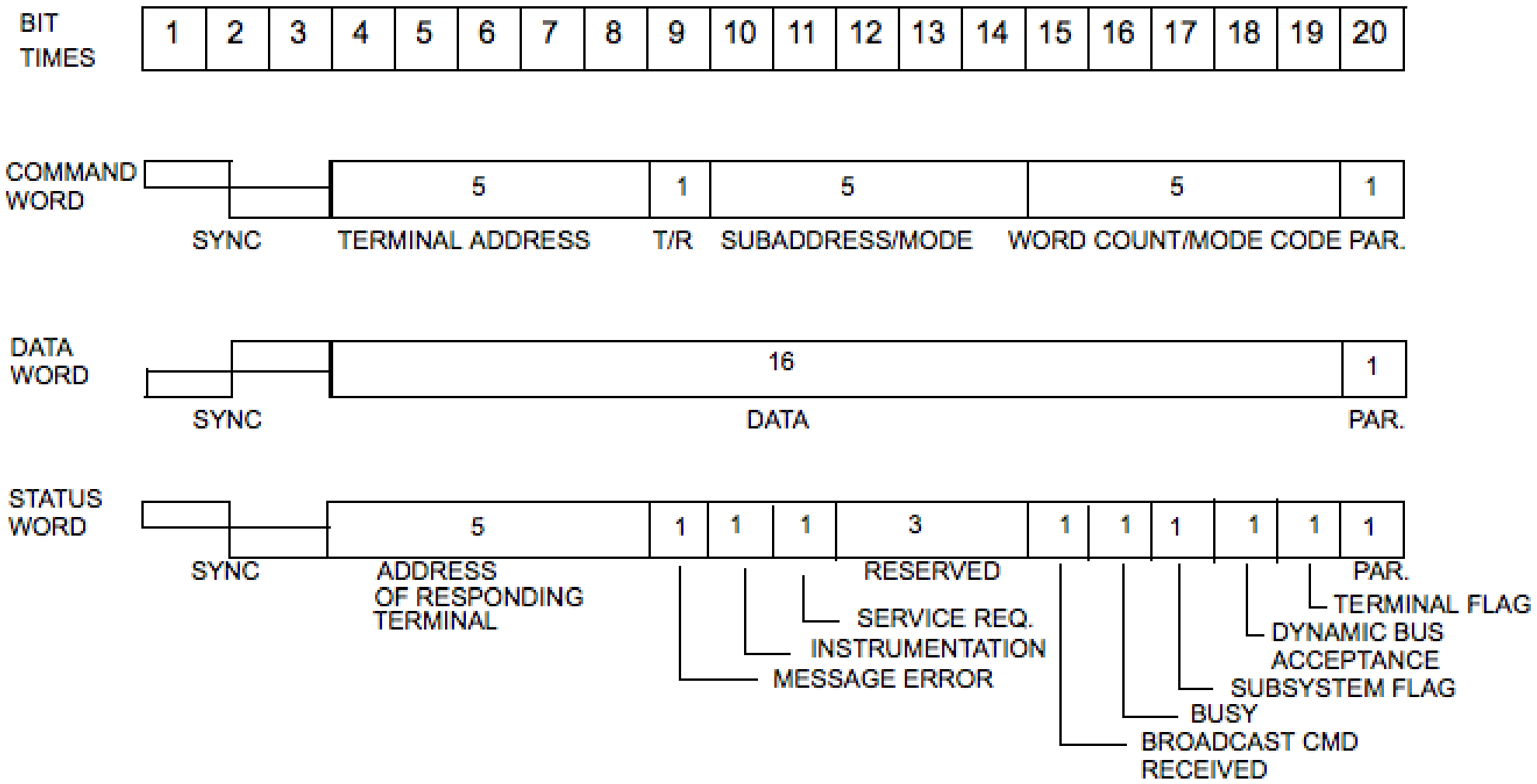
Communication Words



MIL-STD-1553 Tutorial and Reference - Aerospace DAQ, Test, HIL - UEI

Word Types: Command

Command Word Format

Sync.			Remote Terminal Address					T/R	Subaddress / Mode					Data Word Count / Mode Code					Parity
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Command Word Format, Expanded

Section	Bit Count	Description
Sync	1, 2, 3	To aid synchronisation, each word is preceded by a 3 µs sync pulse. The sync pulse is positive for the first half and negative for the second half. The sync pulse is invalid as a Manchester Code.
Source Remote Terminal Address	4, 5, 6, 7, 8	This is the 5 bit address of the Remote Terminal designated to action the command. All but binary address 11111 (decimal 31) identify a unique Remote Terminal. Instead decimal 31 addresses all devices on a "Broadcast Mode", if supported by the bus hardware.
Transmit / Receive	9	Transmit / Receive Value: 0 — The RT is to receive data following the command. Value: 1 — The RT is to transmit data following the command.
Subaddress / Mode	10, 11, 12, 13, 14	Subaddress / Mode Value: 00000 — The Data Word Count should be decoded as Mode Commands. 00001, ..., 11110 — A subaddress that identifies a specific function of the RT, selecting particular data set to be transmitted, or written. 11111 — The Data Word Count should be decoded as Mode Commands.
Data Word Count / Mode Code	15, 16, 17, 18, 19	The Subaddress / Mode value (above) determines how these 5 bits are to be read, this will be either: a) The number of data words to be transmitted or received, where binary 00001 through to 11111 are 1 to 31 decimal, and binary 00000 is actually 32. b) The Optional Mode Code, expanded below.
Parity	20	Used for error detection, this bit is used to ensure that the total number of 1 bits in the previous 16 bits is an odd number. For example the command word 10100 1 00101 01010 has 7 1 bits. This is already an odd number, so the parity bit will be set to 0.

Data Word Format

Sync.			Data																Parity
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

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Command – Mode Codes

The default mode is 00000, but the following codes can be used to send commands.

MIL-STD-1553 Mode Control Code:

T/R	Mode Control	Function	Description	Data Word	Broadcast
1	00000	Dynamic Bus Control	Hands control of the bus to a RT. If the RT declines, the existing bus controller retains control.	No	No
1	00001	Sync.	Reset the internal timer of an RT, initiate a sequence, etc..	No	Yes
1	00010	Transmit Status Word	Causes an RT to transmit the status word associated with the previous valid command so that the BC can confirm receipt.	No	No
1	00011	Initiate Self-Test	Causes an RT to begin a self-test and the transmit it's status.	No	Yes
1	00100	TX Shutdown	Disables the TX associated with the current redundant / backup bus.	No	Yes
1	00101	Override TX	Enables the TX associated with a previously disabled redundant bus.	No	Yes
1	00110	Inhibit Terminal Flag Bit	Sets the T/F bit in the status word to logical zero unless otherwise commanded.	No	Yes
1	00111	Override Inhibit Terminal Flag Bit	Resets the Inhibit Terminal Flag bit.	No	Yes
1	01000	Reset Remote Terminal	Resets an RT to the default power up initialised state.	No	Yes
1	01001	Reserved	Not currently used.	No	TBD
1	.....	Reserved	Not currently used.	No	TBD
1	01111	Reserved	Not currently used.	No	TBD
1	10000	Transmit Vector Word	Cause the RT to transmit a status word and a data word containing service request information.	Yes	No
0	10001	Sync.	The RT shall receive a Command Word followed by a Data Word containing sync. information for the RT.	Yes	Yes
1	10010	Transmit Last Command	Causes an RT to transmit its Status Word.	Yes	No
1	10011	Transmit Built-In-Test Word	Causes an RT to transmit its Status Word followed by a single Data Word containing the Built-In-Test (BIT) data.	Yes	No
0	10100	Selected TX Shutdown	Causes an RT to disable the TX associated with a specific redundant bus.	Yes	Yes
0	10101	Override Selected TX	Enables a TX previously disables by the Selected TX Shutdown command.	Yes	Yes
0 or 1	10110	Reserved	Not currently used.	Yes	TBD
0 or 1	.....	Reserved	Not currently used.	Yes	TBD
0 or 1	11111	Reserved	Not currently used.	Yes	TBD

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Word Types: Data

Section	Bit Count	Description
Sync	1 2 3	To aid synchronisation, each word is preceded by a 3 µs sync pulse. The sync pulse is positive for the first half and negative for the second half. The sync pulse is invalid as a Manchester Code.
Source Remote Terminal Address	4 5 6 7 8	This is the 5 bit address of the Remote Terminal transmitting the Status Word. All but binary address 11111 (decimal 31) identify a unique Remote Terminal. Decimal 31 addresses all devices on a "Broadcast Mode", if supported by the bus hardware.
Message Error	9	Message Error Bit Value: Interpretation 0: No error. 1: One of the following error conditions may have occurred. • A received Data Word contains an error. • Incorrect number of Data Words received. • Unrecognised Command Word received.
Instrumentation	10	Used to aid the distinction between Command and Status Word. Usually set to logical zero.
Service Request	11	Optional bit used to request the Bus Controller to carry out a predefined action.
Reserved	12 13 14	
Broadcast Command	15	Broadcast Command Bit Value: Interpretation 0: Previous command was not a broadcast command. 1: Previous command was a broadcast command.
Busy	16	Busy Bit Value: Interpretation 0: Not busy, or not implemented. 1:
Subsystem Flag	17	Subsystem Flag Bit Value: Interpretation 0: No error, or not implemented. 1: There is a fault on the RT's subsystem.
Dynamic Bus Control Acceptance	18	Dynamic Bus Control Acceptance Bit Value: Interpretation 0: RT has rejected hand over of bus control. 1: RT has accepted hand over of bus control.
Terminal Flag	19	Terminal Flag Bit Value: Interpretation 0: No fault, or not implemented. 1: RT has a fault.
Parity	20	Used for error detection, this bit is used to ensure that the total number of 1 bits in the previous 16 bits is an odd number. For example the command word 10100 1 00101 01010 has 7 1 bits. This is already an odd number, so the parity bit will be set to 0.

Status Word Format

Sync.			Remote Terminal Address					ME	I	SR	Reserved			BC	B	SF	DBC	TF	Parity
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

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Video Review

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Wrapping Up

Last Time:

Learned about the major avionic systems associated with GNC

Today:

Discussed the motivation for the use of Avionics Databuses and Architectures.

Introductory concepts were covered.

Coming up:

Aviation Bus Standards: ARINC 429, ARINC 629, ARINC 664

Questions?

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Module References

- Past Presentations
  - Lecture slides prepared and reused from Dr. Timothy Wilson.
  - Lecture slides prepared and reused from Dr. Eduardo A. Rojas-Nastrucci.

- "Principals of Avionic Systems" by Al Helfrick, 4th edition.

- "Digital Avionics Handbook" by Spitzer, Ferrell, Ferrell.

- "Avionics Full Duplex Ethernet and the Time Sensitive Networking Standard" by Pasquier and Schneele, Airbus.

- MIL-STD-1553, nepp.nasa.gov/docuploads/43745C0A-323E-4346-A434F4342178CD0E/MIL-STD-1553.pdf

- MIL-HDBK-1553, MIL-HDBK-1553 A MULTIPLEX APPLICATIONS

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